

[illegible]

NC pins are intentionally unused. SMD probe connects on the module top.

J6 is a user button. Physical disable is J8; reset can cause a fall.

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3.3 V logic; actuator thresholds and timing remain bench items.

The schematic diagram illustrates the ARM core and its supporting logic. Key components and connections include:

- ARM Core (U5):** A yellow block representing the ARM core, with pins for `CLK`, `CLK_N`, `Q_N`, `ARM_Q`, `ARM_Q_N`, `VCC`, `PRE_N`, and `GND`.
- Latched Enable Loop (J8):** A yellow block with pins `1` and `2`, connected to `LOOP_VDD` and `CLK_N`.
- Reset Logic (U6):** A yellow block labeled `TLVDS6C230PZR` with pins `1` (`RESET_N`), `2` (`CLK_N`), and `3` (`GND`).
- Control Signals:**
 - `ARM_CLK` (pin 1) and `CLK_N` (pin 6) are connected to the ARM core.
 - `ARM_Q` (pin 5) is connected to `ARM_Q_N` (pin 3) via a resistor `R19` (1k).
 - `PRE_N` (pin 7) is connected to `GND` via a resistor `R5` (10k).
 - `CLK_N` (pin 6) is connected to `GND` via a resistor `R7` (10k).
 - `RESET_N` (pin 1) is connected to `GND` via a resistor `R6` (1k).
- Power and Grounding:**
 - `VCC` (pin 4) is connected to `+3V3`.
 - `GND` (pin 8) is connected to `GND`.
 - `CLK_N` (pin 6) is connected to `GND` via a capacitor `C12` (100n).
 - `RESET_N` (pin 1) is connected to `GND` via a capacitor `C6` (100n).
- Other Components:**
 - `D1` (`BATS54H.115`) and `D2` (`BATS54H.115`) are diodes connected to `CLK_N` (pin 6).
 - `D3` (`BATS54H.115`) is a diode connected to `CLK_N` (pin 6).

CLR_N dominates. Re-arm requires a new ARM_CLK edge; this is not a watchdog.

CAM_3V3 powers translator B only. It does not power the motion MCU.

3.3 V logic; actuator thresholds and timing remain bench items.

OE_N = request_N OR ARM_Q_N. Pull-ups disable TX by default.

The diagram shows the timing of IMU module signals relative to the +3V3 supply and the JA BODY I/O pins. The signals are represented by green horizontal lines, and the pins are represented by red vertical lines. The signals are labeled as follows:

- IMU_SCK_M**: IMU_SCK (pin 1)
- IMU_MOSI_M**: IMU_MISO (pin 2)
- IMU_CS_M**: IMU_CS (pin 3)
- IMU_SCK**: IMU_SCK (pin 4)
- IMU_MISO**: IMU_MISO (pin 5)
- IMU_CS**: IMU_CS (pin 6)
- IMU_DRDY**: IMU_DRDY (pin 7)
- END**: END (pin 8)

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- IMU_CS**: IMU_CS (pin 6)
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- END**: END (pin 8)

SPI and DRDY belong to the rigid body IMU, not the head.

Inputs. Scaling is on the power board.

1k / 10nF filters at the ADC inputs. Scaling is on the power board.